

Reinforcement Learning

Temporal-Difference Learning

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Temporal-Difference (TD) Learning

- Combination of Monte Carlo (MC) and dynamic programming (DP)
 - ▶ Like MC, directly learn from raw experience
 - ▶ Like DP, update estimates based on other learned estimates (bootstrap)
- For control, DP, TD, and MC methods all use some variation of generalized policy iteration (GPI).
- The difference primarily lies in the prediction problem.

TD Prediction

- Update rule for (every-visit) Monte Carlo

$$V(S_t) \leftarrow V(S_t) + \alpha [G_t - V(S_t)]$$

where $G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots \gamma^{T-t-1} R_T$

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- TD(0) (one-step TD) update rule

$$V(S_t) \leftarrow V(S_t) + \alpha [R_{t+1} + \gamma V(S_{t+1}) - V(S_t)]$$

- TD methods need to wait only until the next time step

Comparing Targets of MC and TD

- Prediction: learn v_π for a given policy π

$$\begin{aligned}v_\pi(s) &:= \mathbb{E}_\pi[G_t \mid S_t = s] \\&= \mathbb{E}_\pi[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots \mid S_t = s] \\&= \mathbb{E}_\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s] \\&= \mathbb{E}_\pi[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t = s]\end{aligned}$$

- MC target: G_t vs. TD target: $R_{t+1} + \gamma V(S_{t+1})$

TD(0) Pseudocode for Prediction

Tabular TD(0) for estimating v_π

Input: the policy π to be evaluated

Algorithm parameter: step size $\alpha \in (0, 1]$

Initialize $V(s)$, for all $s \in \mathcal{S}^+$, arbitrarily except that $V(\text{terminal}) = 0$

Loop for each episode:

 Initialize S

 Loop for each step of episode:

$A \leftarrow$ action given by π for S

 Take action A , observe R, S'

$V(S) \leftarrow V(S) + \alpha[R + \gamma V(S') - V(S)]$

$S \leftarrow S'$

 until S is terminal

TD Error

- **TD error**: difference between current estimate $V(S_t)$ and TD target $R_{t+1} + \gamma V(S_{t+1})$

$$\delta_t := R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$$

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- Error in the estimate made at time t
 - ▶ Depends on the next state and next reward, it is not actually available until one time step later
 - ▶ Error in $V(S_t)$, available at time $t + 1$

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 - ▶ Depends on the next state and next reward, it is not actually available until one time step later
 - ▶ Error in $V(S_t)$, available at time $t + 1$
- If array V does not change during the episode, TD error is equivalent to MC error

- ▶ $G_t - V(S_t) = \sum_{k=t}^{T-1} \gamma^{k-t} \delta_k$ (See pg.121 of the textbook for derivation)

Leave

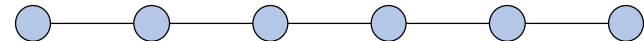
Reach Car

Exit highway

2ndary road

Home street

Arrive home

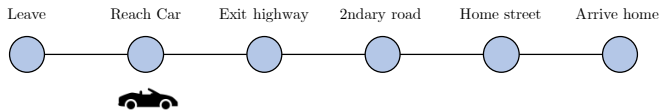


Elapsed
Time

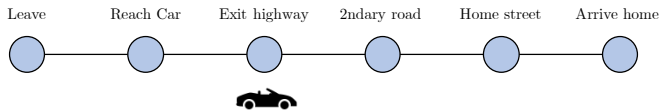
0 mins

Predicted
Time to Go

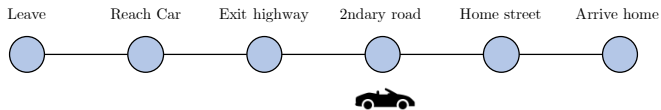
30 mins



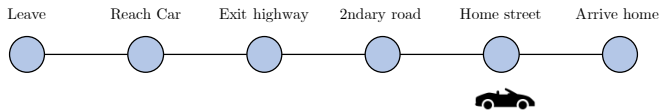
Elapsed Time	0 mins	5 mins
Predicted Time to Go	30 mins	35 mins



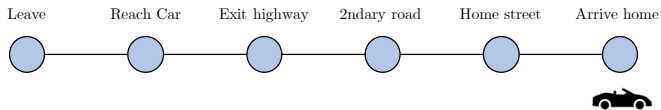
Elapsed Time	0 mins	5 mins	20 mins
Predicted Time to Go	30 mins	35 mins	15 mins



Elapsed Time	0 mins	5 mins	20 mins	30 mins	
Predicted Time to Go	30 mins	35 mins	15 mins	10 mins	

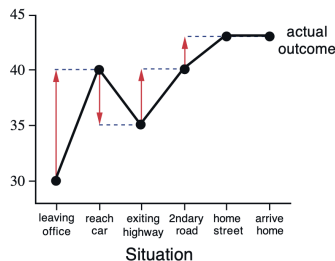
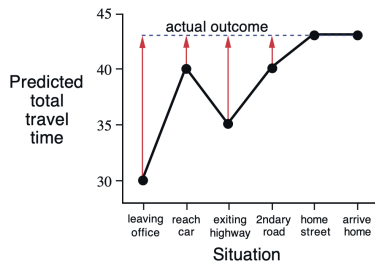


Elapsed Time	0 mins	5 mins	20 mins	30 mins	40 mins	
Predicted Time to Go	30 mins	35 mins	15 mins	10 mins	3 mins	



Elapsed Time	0 mins	5 mins	20 mins	30 mins	40 mins	43 mins
Predicted Time to Go	30 mins	35 mins	15 mins	10 mins	3 mins	0 mins

MC vs TD Updates



- Monte Carlo update

$$V(S_t) \leftarrow V(S_t) + \alpha [G_t - V(S_t)]$$

- TD(0) update

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Advantages of TD Prediction Methods

- MC must wait until the end of an episode
- TD methods only need wait only one time step

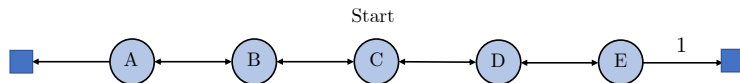
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- (Case 1) Very long episodes, so that delaying all learning until the end of the episode is too slow
- (Case 2) Continuing tasks and have no episodes at all.

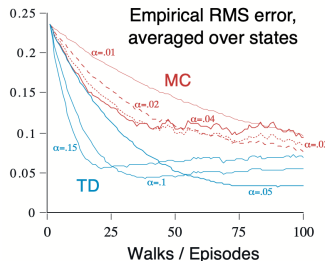
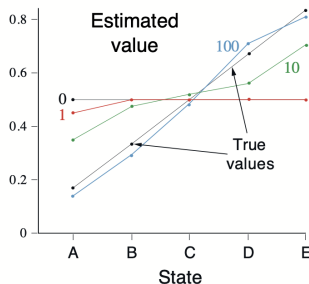
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- (Case 2) Continuing tasks and have no episodes at all.
- But, does TD still guarantee convergence to the correct answer?

Chain MDP



- True values $V(A) = \frac{1}{6}$, $V(B) = \frac{2}{6}$, $V(C) = \frac{3}{6}$, $V(D) = \frac{4}{6}$, $V(E) = \frac{5}{6}$



Next class: TD Learning Control
(Chapter 6 of Sutton & Barto)